PHARMACOVIGILANCE · TASK 4

Causality, Seriousness and Expectedness Assessment

Student Name: Tanuja Patil

Task: Task 4 · Causality, Seriousness & Expectedness
Product: Velunex
Assessment method: WHO-UMC framework / structured clinical reasoning

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1. Assessment Matrix

Case ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
PV-201	Dizziness began 2 hours after first use; resolved same day; no treatment/hospitalization	**Possible**	**Non-serious** – no hospitalization, life-threatening event, death, disability or other serious outcome reported	**Mild**	**To be verified against Velunex reference**	Strong temporal relationship because dizziness began 2 hours after exposure. However, alternative causes, concomitant products and other clinical information are unavailable. Same-day spontaneous resolution supports a transient event but does not establish causality.

Case ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
PV-202	Appendicitis occurred 3 weeks after first use and required hospitalization; medical history incomplete	**Unlikely / Unassessable**	**Serious – hospitalization criterion**	**Not supplied**	**To be verified against Velunex reference**	Hospitalization makes the event serious. The 3-week interval provides weaker temporal support, and incomplete medical history prevents adequate assessment of alternative causes. Further clinical information is needed.
PV-203	Mild rash began on day 4; improved after stopping; no rechallenge	**Probable**	**Non-serious** – no serious outcome or hospitalization reported	**Mild**	**To be verified against Velunex reference**	The onset after exposure and improvement after discontinuation provide supportive temporal and dechallenge evidence. However, rechallenge was not performed and alternative causes/concomitant

Case ID	Reported adverse event	Causality assessment	Seriousness & criterion	Severity	Expectedness	Overall rationale
						products are unknown, so causality cannot be considered definite.
PV-204	Loss of consciousness with emergency admission on day 2; concomitant products and test results unknown	**Unassessable / Possible**	**Serious – emergency admission; potentially medically important/life-threatening depending on clinical documentation**	**Severe**	**To be verified against Velunex reference**	The event occurred after exposure and required emergency admission, but the absence of diagnostic results and concomitant-product information prevents reliable exclusion of alternative causes. Additional clinical information is essential.

2. Detailed Case Assessments

PV-201 – Dizziness

Case information:
Dizziness began two hours after first use of Velunex. It resolved the same day. No treatment or hospitalization was required.

Assessment

- **Causality:** Possible
- **Seriousness:** Non-serious
- **Seriousness criterion:** None identified
- **Severity:** Mild
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Not specifically reported; event resolved spontaneously
- **Rechallenge:** Not reported
- **Alternative causes:** Unknown

Rationale

The two-hour interval between Velunex exposure and onset of dizziness provides a plausible temporal relationship. However, there is insufficient information about concomitant medicines, medical history, alternative causes, or rechallenge. Therefore, the available information supports possible causality, but not a definite causal relationship.

Missing information / Follow-up

1. What was the dose and route of Velunex?
2. Was the patient taking any concomitant medicines?
3. Did the patient have a history of dizziness, hypotension or vestibular disease?
4. Were vital signs available?
5. Did dizziness recur after subsequent exposure?

PV-202 – Appendicitis

Case information:
Appendicitis occurred three weeks after first use of Velunex and required hospitalization. Medical history is incomplete.

Assessment

- **Causality:** Unlikely / Unassessable
- **Seriousness:** Serious
- **Seriousness criterion:** Hospitalization
- **Severity:** Not supplied

- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Not applicable/not reported
- **Rechallenge:** Not reported
- **Alternative causes:** Cannot be adequately assessed because medical history is incomplete

Rationale

The hospitalization means the case meets a **seriousness criterion**. However, appendicitis occurred three weeks after first exposure, which provides relatively weak temporal evidence for a drug-related reaction. In addition, the medical history is incomplete. Therefore, a causal relationship with Velunex cannot be established from the available information.

Missing information / Follow-up

1. Was appendicitis confirmed clinically or surgically?
2. What was the patient's previous gastrointestinal/medical history?
3. Were any concomitant medicines being taken?
4. Were laboratory/imaging results available?
5. What treatment was given?
6. What was the final outcome?

PV-203 – Mild Rash

Case information:
A mild rash began on day four. It improved after stopping Velunex. No rechallenge was performed.

Assessment

- **Causality:** Probable
- **Seriousness:** Non-serious
- **Seriousness criterion:** None identified
- **Severity:** Mild
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Positive

- **Rechallenge:** Not performed
- **Alternative causes:** Unknown

Rationale

The rash developed after exposure to Velunex and improved after discontinuation, providing both a plausible temporal relationship and positive dechallenge. These findings support a **probable association**. However, because no rechallenge was performed and information about concomitant products and alternative causes is unavailable, the relationship should not be classified as definite.

Missing information / Follow-up

1. What was the exact type and distribution of the rash?
2. Was there itching, swelling, fever or mucosal involvement?
3. Were any other medicines/products used?
4. Did the rash completely resolve?
5. Was any treatment required?
6. Did the patient have a previous history of allergy?

PV-204 – Loss of Consciousness

Case information:
Loss of consciousness occurred on day two and resulted in emergency admission. Concomitant products and test results are unknown.

Assessment

- **Causality:** Unassessable / Possible
- **Seriousness:** Serious
- **Seriousness criterion:** Emergency hospitalization; potentially medically important/life-threatening depending on documented clinical circumstances
- **Severity:** Severe
- **Expectedness:** To be verified against the supplied Velunex reference
- **Dechallenge:** Unknown
- **Rechallenge:** Unknown
- **Alternative causes:** Cannot be assessed adequately

Rationale

Loss of consciousness requiring emergency admission represents a clinically significant event and meets the seriousness criterion based on hospitalization. The event occurred two days after exposure, which provides some temporal support. However, diagnostic test results and concomitant-product information are unavailable. Syncope/loss of consciousness can have many alternative causes, so the available evidence is insufficient to establish a stronger causal relationship.

Missing information / Follow-up

1. What was the duration of unconsciousness?
2. Were blood glucose, blood pressure and ECG results available?
3. Were neurological investigations performed?
4. Were any concomitant medicines/products being used?
5. Was there a previous history of syncope, seizures or cardiovascular disease?
6. What was the final diagnosis?
7. Did the patient recover completely?
8. Was hospitalization prolonged?

Case	**Event**	**Causality**	**Serious?**	**Seriousness criterion**	**Severity**	**Expectedness***
PV-201	Dizziness	**Possible**	**No**	None	Mild	**Verify reference**
PV-202	Appendicitis	**Unlikely / Unassessable**	**Yes**	Hospitalization	Not supplied	**Verify reference**
PV-203	Rash	**Probable**	**No**	None	Mild	**Verify reference**
PV-204	Loss of consciousness	**Unassessable / Possible**	**Yes**	Emergency admission / medically important	Severe	**Verify reference**

4. Overall Conclusion

The four cases demonstrate why causality, seriousness, severity and expectedness must be assessed separately.

- **PV-201:** temporally associated dizziness, but insufficient evidence for a definite relationship.
- **PV-202:** serious because hospitalization occurred; causal association is weak/uncertain because of the three-week interval and incomplete medical history.
- **PV-203:** strongest causal evidence among the four cases because of temporal association and positive dechallenge; classified as probable.
- **PV-204:** serious because emergency admission occurred, but causality remains uncertain because important diagnostic and concomitant-product information is missing.

No case should be classified as definitely drug-related based solely on the information supplied.